

Name: Carlos Bracho
Professor: Jeff Sanford

CS 499 3-2 Milestone Two Narrative Enhancement One Software Design and Engineering

Artifact Description

The artifact I selected for this milestone and for all of 3 categories is my inventory application, which I am currently developing as part of my Computer Science Capstone course. This project is a full-stack application built using the MERN stack, including MongoDB, Express, React, and Node.js. As a reminder, I am basically recreating my previous Android App, which was created using Kotlin and had a database that was stored directly on the android device. I am moving that app to the web with better Frontend, Backend and a database that is stored in the cloud.

Originally, I had experience with other technologies, but I decided to restart this project from scratch and use React as a challenge to myself, so I can learn something that I consider to be a more modern and widely used framework. I created a new GitHub repository and structured the project into separate frontend and backend folders to improve organization and scalability. This is the link to the repository: <https://github.com/BrachoDev/inventoryApp>

The application allows users to manage inventory items by performing CRUD operations (create, read, update, and delete products). For this milestone, I focused on building and enhancing the backend functionality.

Justification for Inclusion

I selected this artifact for my ePortfolio because it demonstrates my ability to design and develop a full-stack application using modern tools and best practices in software engineering. One of the main reasons I chose this project is because of the complexity involved in setting up

the backend. I configured a Node.js server using Express, connected it to a MongoDB Atlas database, and managed environment variables securely using dotenv. I also structured the backend by creating a models folder and defining schemas using Mongoose. This reflects real-world backend development practices.

A key component that showcases my skills is the implementation of full CRUD functionality:

- A POST method to create new products
- A GET method to retrieve all products
- A DELETE method to remove products
- A PUT method to update existing products

The PUT method, in particular, was part of my planned enhancement and required debugging and testing using Postman. This process helped me better understand how APIs handle updates and how to properly structure requests and responses. Additionally, I began improving the design of my application by planning to modularize my server code into separate routes and expand the database by adding an user collection to start handling user related data. These improvements show my ability to think ahead about scalability and maintainability.

Course Outcomes Reflection

Through this enhancement, I made progress toward several Computer Science program outcomes. First, I worked on designing and evaluating solutions by implementing a fully functional backend system that connects to a database and supports CRUD operations. This required me to think through how data flows between the client, server, and database. Second, I demonstrated my ability to use modern tools and techniques in software development. I used technologies such as Node.js, Express, MongoDB Atlas, and Mongoose, which are commonly

used in the Computer Science industry. Finally, I applied software engineering principles by organizing my project into separate folders, using environment variables for security, and planning for modular code structure. These steps show an understanding of maintainability and scalability in application design.

At this stage, I believe I am making solid progress toward these outcomes and plan to continue improving areas such as security and user authentication in future enhancements.

Reflection on the Enhancement Process

Throughout this process, I learned a lot about backend development and the challenges that come with setting up a full-stack application. One of the first challenges I faced was dealing with outdated versions of Node.js and npm. This caused errors when initializing the project, and I had to troubleshoot and update my environment before continuing. This experience taught me how important it is to maintain an up-to-date development environment. Another challenge was connecting to MongoDB Atlas and configuring the connection string correctly using environment variables. Debugging this helped me understand how backend applications securely connect to databases and how small configuration issues can cause major problems.

I also learned a lot while implementing and testing my API endpoints. The PUT method was especially challenging, as I had to debug issues using Postman and ensure that the request body and database updates were working correctly. This helped me better understand how APIs function and how to handle data updates properly. I am enjoying this part of my project because it has been helping me with becoming more confident in backend development and gave me practical experience with tools and workflows used currently. It also reinforced the importance of debugging, testing, and planning for future improvements when designing software.